Digital Twin Anchored Generation: Adaptive Questioning for Directed Movement in Opinion Space

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***Abstract*—Can the selection of survey questions move a respondent's model-inferred ideology coordinate without controlling the answers? Digital Twin Anchored Generation (DTAG) uses Large Science Model (LSM) conditionals to select items maximizing expected displacement in a pole-relative predictive phenotype. In 960 GSS 2024 twenty-question simulations across six personas, directional policies shifted the index without asking any of the 27 pole-defining items; 2,880 fixed-set replays assessed effects of question order. The completed prospective Prolific study comprises 71 calibrated sessions: 21 Left, 38 Right, 7 Neutral, and 5 Random. At the end of the adaptive sequence, 20/21 Left-push participants (95.2%) moved left, compared with 21/38 Right-push participants (55.3%) moving right. Rightward movement depended strongly on the initial inferred position: 12/29 initially Left-leaning participants (41.4%) and 9/9 initially Right-leaning participants (100%) moved right. Among the 17 initially Left-leaning Right-push nonresponders, all were flagged as resistant to baseline-relative rightward movement after four adaptive questions (sensitivity 100%, accuracy 27/29, 93.1%). The observed changes concern conditional survey-response phenotypes rather than independently verified changes in political beliefs.**

***Index Terms*—Predictive phenotypes, large science models, conditional response modeling, digital twins, survey methodology, opinion dynamics, adaptive questioning, generative simulation.**

I. Introduction

Survey questions are usually treated as measurements, but each answer also changes predictions about unanswered items. We ask whether choosing *which questions to pose* can steer a respondent's inferred ideological position. If so, adaptive surveys do more than collect information: their selection policies shape the modeled opinion profile.

Large Science Models (LSMs) provide a new framework for generative digital twins of multivariate systems. They learn interdependent conditional distributions, infer unobserved variables from partial evidence, and compare states through their predictive consequences. Related digital-twin applications include infant microbiome development [1] and influenza evolution through Emergenet [2]. DTAG applies an LSM to survey answers, defining a *predictive phenotype* over unasked items and its position relative to opposing reference opinion profiles. It selects questions to maximize expected signed displacement while respondents supply the answers.

We test this with 960 native GSS 2024 trajectories across six personas (excluding all pole-defining items), 2,880 fixed-set order replays, and 71 completed calibrated human sessions recruited through Prolific. The human results additionally show that baseline-relative Right-push nonresponse among initially Left-leaning participants is identifiable after four adaptive questions.

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II. Related Work

Attitude dependencies need not align on one ideological axis [3], [4]. Partisan sorting and social influence shape these dependencies [5], [6]; cognitive consistency, wording, and order can affect reported attitudes [7]–[9]. This motivates separating information revealed by a question from any change induced by asking it.

Opinion-dynamics models often use low-dimensional transitions [10]–[12], while experiments on exposure and contact report heterogeneous effects [13]–[15]. Generative agents and prompted survey personas can produce plausible responses [16], [17]. DTAG instead combines learned high-dimensional survey conditionals with an explicit question-selection objective.

III. Large Science Models and the Predictive Phenotype

A. Conditional dependency model

Let $X = (X_1, \ldots, X_n)$ be categorical survey responses, with alphabets $\mathcal{A}_i$ and survey context c. An LSM comprises conditionals

$$\Phi_c = \{\phi_{i,c}(a \mid x_{-i}) : i = 1, \ldots, n\}, \tag{1}$$

estimated by conditional inference trees [1], [18], [19]. Their dependencies are predictive, not causal; the conditionals need not form a coherent joint law [20].

A respondent state $s = \{X_j = a_j : j \in S\}$ holds observed responses. Unanchored variables remain missing in $\widetilde{x}_{-v}(s)$; each native tree averages over its missing branches. For unasked $v \notin S$,

$$p_\Phi(a \mid s, v) = \phi_{v,c}(a \mid \widetilde{x}_{-v}(s)). \tag{2}$$

The vector

$$\mathcal{B}_\Phi(s) = \big(p_\Phi(\cdot \mid s, v)\big)_{v \notin S} \tag{3}$$

is the *predictive phenotype*: conditional distributions and uncertainty for unasked responses.

B. Geometry as predictive equivalence

Conditional predictions induce a distance between complete or partial states x, y:

$$\theta_\Phi(x, y) = \frac{1}{n} \sum_{i=1}^{n} \sqrt{D_{\mathrm{JS}}(\phi_{i,c}(\cdot \mid x_{-i}), \phi_{i,c}(\cdot \mid y_{-i}))}. \tag{4}$$

Here D_{JS} denotes Jensen–Shannon divergence. The resulting predictive pseudometric distinguishes states by their conditional consequences.

Prior LSM reconstruction reduced mean distance distortion by 54.9% (GSS) and 60.9% (Eurobarometer) with 80% of responses masked [19].

For GSS models with registered left and right reference states s_L, s_R, DTAG reports

$$I_\Phi(s) = \frac{\theta_\Phi(s_L, s) - \theta_\Phi(s_R, s)}{\theta_\Phi(s_L, s_R)}. \tag{5}$$

The reference poles specify opposing responses on 27 GSS 2024 items covering policy, civil liberties, spending, and religion (Table I) [19]. Other coordinates are unspecified; I_Φ measures predictive position rather than self-reported ideology.

IV. Adaptive Question Selection

A semantic interface can map a natural-language question q to a survey item $v \in M(q)$ and verbalize its answer. The native experiments use item identifiers directly without the LLM layer. For a selected question, DTAG samples a response and updates the predictive phenotype:

$$A_t \sim p_\Phi(\cdot \mid s_t, v_t), \tag{6}$$

$$s_{t+1} = s_t \cup \{v_t = A_t\}, \tag{7}$$

$$\mathcal{B}_{t+1} = \mathcal{B}_\Phi(s_{t+1}). \tag{8}$$

The initial persona anchors remain fixed. The optimizer chooses the question, not its answer (Fig. 1).

For candidate item v and answer a, define

$$\Delta_v(a; s) = I_\Phi(s \cup \{v = a\}) - I_\Phi(s), \tag{9}$$

$$\mu_v(s) = \sum_{a \in \mathcal{A}_v} p_\Phi(a \mid s, v)\Delta_v(a; s), \tag{10}$$

$$\sigma_v^2(s) = \sum_a p_\Phi(a \mid s, v)(\Delta_v(a; s) - \mu_v(s))^2. \tag{11}$$

Right- and left-seeking policies maximize and minimize $\mu_v(s)$; neutral minimizes $|\mu_v(s)|$, and random samples an eligible item. By construction,

$$\mathbb{E}[I_\Phi(s_{t+1}) - I_\Phi(s_t) \mid s_t, v_t] = \mu_{v_t}(s_t).$$

The accumulated displacement depends on successive selected items and sampled answers.

The effect on an unasked item j after observing (v, a) is

$$R_j(s; v, a) = D_{\mathrm{JS}}\big(p_\Phi(\cdot \mid s, j), p_\Phi(\cdot \mid s \cup \{v = a\}, j)\big). \tag{12}$$

A weighted sum $\mathcal{P}_\Phi(s; v, a) = \sum_{j \notin S \cup \{v\}} w_j R_j(s; v, a)$ measures redistribution across the remaining items. Unlike text retrieval [21], DTAG maintains an explicit conditional response state. The public implementation includes a web application (https://github.com/zeroknowledgediscovery/DTAG/tree/main/webapp).

V. Refinement and Possible Response Effects

Under *phenotypic refinement*, successive answers restrict predictions for unasked items. For held-out responses Y, the expected information from asking v is

$$\mathcal{G}(v \mid s) = H(Y \mid s) - \mathbb{E}_{A_v \mid s}[H(Y \mid s, v = A_v)]. \tag{13}$$

Under a coherent joint model this is conditional mutual information, distinct from directional displacement $\mu_v(s)$. Held-out response prediction evaluates refinement directly.

A. Possible response effects.

Questioning a person may change later reports. A response–transition kernel separates an observed answer from a possible latent-state update:

$$K_q(a, z' \mid z) = O_q(a \mid z)\, T_q(z' \mid z, a). \tag{14}$$

Passive observation has $T_q(z' \mid z, a) = \delta_z(z')$; shifts in salience or belief can induce nontrivial transitions. A common held-out battery and delayed measurement can investigate these effects.

B. Order dependence.

For fixed realized answers, the terminal index is order-independent; sequentially generated answers need not be. A coherent joint distribution satisfies

$$P(a \mid s, v)P(b \mid s, v = a, w) = P(b \mid s, w)P(a \mid s, w = b, v). \tag{15}$$

Separately fitted LSM conditionals may violate this identity [20]. Fixed-set replays evaluate its consequences for native response sequences.

VI. Native GSS 2024 Computational Experiments

A. Adaptive questioning from pinned persona states

We fixed six initial GSS 2024 persona states (Table II) and generated 40 independent response sequences per persona and policy (right, left, neutral, random). Each sequence comprised 20 questions with responses sampled from the native LSM; no LLM mapping or verbalization was used. *All 27 pole-defining items were excluded from selection in every policy* (Table I). Thus 960 adaptive trajectories contain 19,200 non-pole responses; repetitions share initial anchors but have independent sampled answers.

Across all six personas, right- and left-seeking policies moved the index in their intended directions (Fig. 1). As pole-defining questions were excluded, the movement results from conditioning on other survey items, not directly asking those used to define the reference index. Neutral and random policies provide comparisons.

B. Question selection as observed paths

Question identities are summarized in Table III. The 18 displayed heatmaps contain 161 distinct items across right, left, and neutral selection (Fig. 2). A shared categorical ordering approximately minimizes

$$\mathcal{C}(\pi) = \sum_{p,u,t} \sum_{r<r'} \big|\pi(v_{purt}) - \pi(v_{pur't})\big|, \tag{16}$$

Here u indexes policies. Heatmap colors identify items, not ideological position; 67 further items occurred only in random selection.

TABLE I

GSS 2024 reference poles. TWENTY-SEVEN PAIRED RESPONSES DEFINE s_L, s_R, WITH $I_\Phi = -1, +1$; ALL OTHER COORDINATES ARE UNSPECIFIED.

Item	Survey topic	Left	Right
abany	Abortion on request	yes	no
abdefect	Abortion: fetal defect	yes	no
abnomore	Abortion: no more children	yes	no
abpoor	Abortion: low income	yes	no
abrape	Abortion after rape	yes	no
absingle	Abortion: unmarried	yes	no
bible	Bible interpretation	ancient book	inspired word
colcom	Communist as college teacher	not fired	yes, fired
confed	Trust in federal government	a great deal	hardly any
conlabor	Trust in organized labor	a great deal	hardly any
grass	Marijuana legalization	should be legal	should not be legal
gunlaw	Permit to purchase gun	favor	oppose
libcom	Communist books in library	not remove	remove
libmslm	Muslim cleric books in library	not remove	remove

Item	Survey topic	Left	Right
natarms	Military spending	too much	about right
natenvir	Environmental spending	too little	about right
natfare	Welfare spending	too little	too much
natsoc	Social Security spending	too little	about right
owngun	Guns in household	no	yes
pillok	Contraception access	strongly agree	disagree
polabuse	Police strike insulting citizen	yes	no
pray	Frequency of prayer	never	several times a day
prayer	Bible prayer in public schools	approve	disapprove
religen	Personal religiosity	no religion	strong
rowngun	Respondent owns a gun	no	yes
shotgun	Owns a shotgun	no	yes
spkcom	Communist public speech	yes, allowed to speak	not allowed

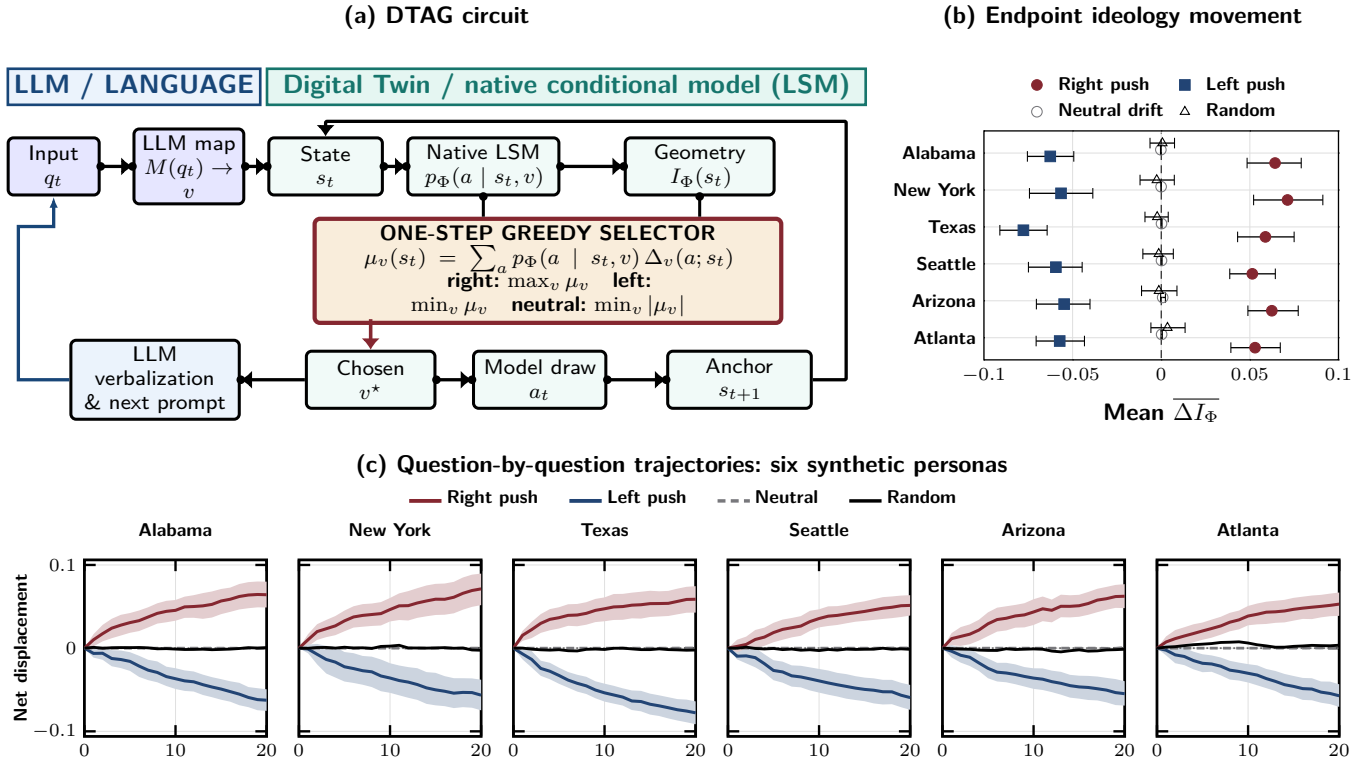


Fig. 1. **DTAG architecture and native adaptive results.** (a) LSM-anchored responses and directional question selection; language mapping is a separate interface. (b) Mean endpoint displacement across six personas with 95% bootstrap intervals. (c) Mean 20-question trajectories over 40 repetitions per persona and policy, with pointwise 95% intervals (right: red; left: blue; neutral: dashed gray; random: black).

C. Fixed-set replay: robustness to question ordering

For each persona and right/left/neutral policy, we froze the 20-item sequence from adaptive replicate zero, giving 18 fixed sets. Each set was replayed in original, reversed, and two shuffled orders with 40 fresh sampled responses per order (2,880 runs). Matched question-specific random variates reduced contrast variance.

The direction persisted across replays (Fig. 2): right-derived sets yielded mean displacement +0.0381 (original) and +0.0377 (reversed); left-derived sets yielded −0.0399 and −0.0410. In both orders, all six persona-specific 95% bootstrap intervals excluded zero in the prescribed direction. Shuffles followed the same pattern; neutral sets remained near zero.

All 54 pointwise bootstrap intervals for paired mean between-order contrasts included zero; no systematic mean order effect was detected. Selected question sets therefore retained directional displacement without continuous reoptimization.

In historical ten-question runs, baseline indices differed across environments despite matched persona, map, and pole hashes; native binaries were not compared. Their absolute indices are not combined with the present experiment.

TABLE II
PINNED GSS 2024 SYNTHETIC PERSONAS AND BASELINE POLE-RELATIVE INDEX.

Persona	Pinned description	Initial $I_\Phi(s_0)$
Alabama	45-year-old rural Alabama white man; married, two children; full-time farmer and veteran; high school; conservative, strong Republican; Protestant, weekly attendance; daily newspaper.	0.075706
New York	22-year-old New York City white woman; unmarried, no children; full-time retail worker, nonveteran; bachelor's degree; extremely liberal, strong Democrat; nonreligious; daily newspaper.	−0.163807
Texas	61-year-old small-town Texas white woman; married, three children; homemaker, high school; extremely conservative, strong Republican; Protestant, attends services several times weekly; daily newspaper.	0.045379
Seattle	34-year-old Seattle white man; unmarried, no children; employed full time, graduate degree; liberal, weak Democrat; nonreligious; newspaper several times weekly.	−0.119639
Arizona	39-year-old Mexican American man in an Arizona city; married, three children; employed full time, high school; conservative, weak Republican; Catholic, nearly weekly attendance.	−0.020437
Atlanta	47-year-old Atlanta Black woman; divorced, two children; employed full time, bachelor's degree; liberal, strong Democrat; Protestant, weekly attendance; daily newspaper.	−0.030807

Positive values lie nearer the right reference pole; negative values nearer the left.

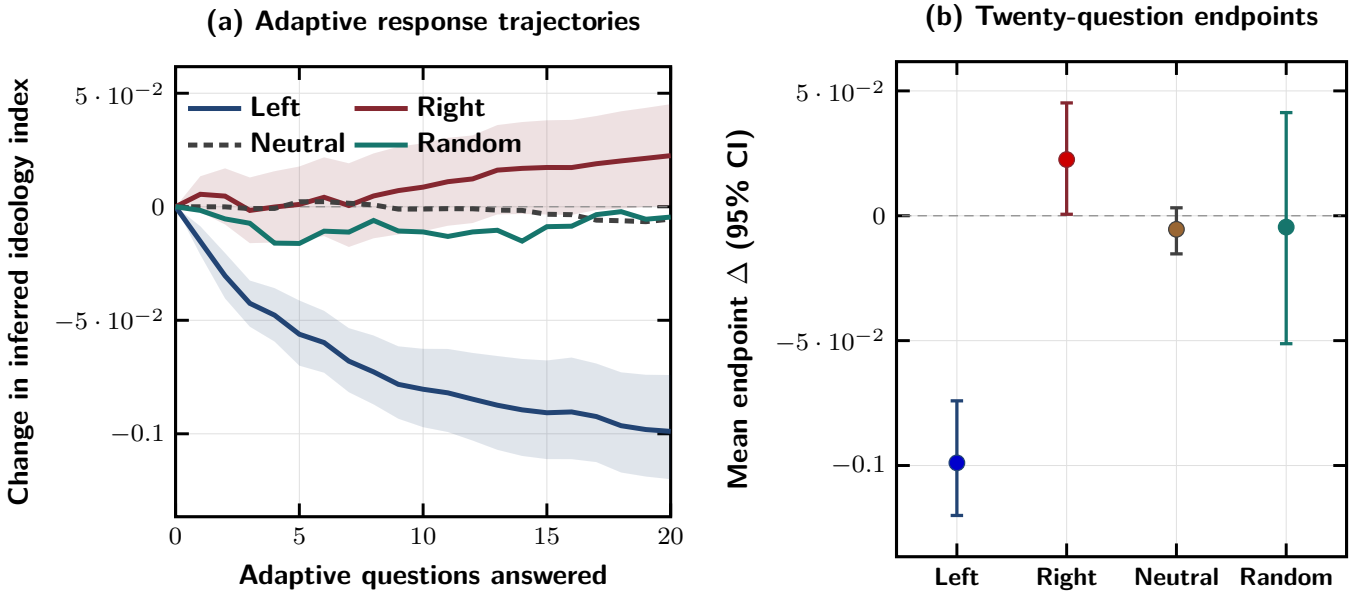


Fig. 2. **Selected questions and fixed-set replays.** Left: question-ID heatmaps for six personas and three selection policies (40 runs each); color denotes item identity. Right: mean index displacement and 95% bootstrap intervals for 20-item sets replayed in original, reversed, and two shuffled orders. Right- and left-derived sets retain their direction; neutral sets remain near zero.

TABLE III
QUESTION SELECTION IN THE NATIVE GSS 2024 EXPERIMENT.

Selection policy	Distinct	Selections	Pole overlap
Right-seeking	100	4,800	0
Left-seeking	75	4,800	0
Neutral	43	4,800	0
Three-policy union	161	14,400	0
All four policies	228	19,200	0

The random policy contributed 4,800 selections; 67 of the 228 distinct items occurred only under random selection.

VII. HUMAN RESPONSE STUDY

The human experiment measures the realized response of the model-inferred ideology index under adaptively selected questions. The 71 completed calibrated sessions and the four-question resistance result are reported below; the experimental protocol and LSM geometry are unchanged.

A. *Protocol and outcome*

We recruited participants through Prolific for the prospective DTAG human-response experiment. Each calibrated session comprised six baseline questions, twenty adaptive questions, and five interleaved attention checks. Participants answered in natural language; accepted native survey-variable assignments updated the LSM state. Four question-selection policies were used: Left, Right, Neutral, and Random. The original study randomized policies; subsequent cohorts used fixed Left or Right assignments. Pooling across recruitment waves is descriptive, not a randomized comparison between waves.

Let s_0 be the state after the six baseline questions and s_t the state following adaptive answer t. The index trajectory

and baseline-relative displacement are

$$\Delta I_\Phi(t) = I_\Phi(s_t) - I_\Phi(s_0), \qquad t = 0, \ldots, 20. \tag{17}$$

A directional success at the end of the adaptive sequence is $\Delta I_\Phi(20) < 0$ for Left and $\Delta I_\Phi(20) > 0$ for Right. The index is the LSM-inferred survey-response phenotype, not a measured change in underlying political beliefs.

B. Completed sessions and directional response

The calibrated study contains 71 participants who completed the 20 adaptive questions: 25 from the original four-policy experiment, 15 from the fixed-Left follow-up, and 31 from the fixed-Right follow-up. The assigned-policy totals are Left ($n = 21$), Right ($n = 38$), Neutral ($n = 7$), and Random ($n = 5$). The separately recruited zero-calibration Right cohort is not combined with these participants, since it does not have an individually measured six-question baseline.

The archived results show leftward displacement in 20/21 Left-push participants (95.2%) and rightward displacement in 21/38 Right-push participants (55.3%). Baseline position strongly distinguishes the observed Right-push response: 12/29 initially Left-leaning participants (41.4%) moved right, compared with 9/9 initially Right-leaning participants (100%). For Left push, 16/17 initially Left-leaning participants (94.1%) and 4/4 initially Right-leaning participants (100%) moved left.

C. Early identification of Right-push resistance

Right-directed questioning produced a weaker response among initially Left-leaning participants. We therefore examined whether resistance to rightward displacement could be identified from the first four adaptive responses. This analysis was restricted to the 29 Right-push participants whose baseline inferred ideology index was negative. A participant was flagged as resistant if their inferred index had not increased relative to their own baseline after four questions. We evaluated that early classification against whether the index had moved rightward from baseline at the end of the full 20-question sequence.

Seventeen of these 29 participants showed no net rightward displacement at the end of the study. All 17 had been identified by the four-question criterion. The criterion also flagged two participants who ultimately moved rightward. It therefore identified all observed resistant participants (17/17, 100% sensitivity), with 89.5% precision (17/19), 83.3% specificity (10/12), and 93.1% overall classification accuracy (27/29).

Rightward displacement or correctly identified endpoint nonresponse accounts for all 29 initially Left-leaning Right-push participants (12 and 17, respectively); including initially Right-leaning participants, the combined outcome is 38/38. This is a composite of successful directional response and correct early identification of nonresponse, *not* a 38/38 directional intervention success rate.

D. Interpretation

The completed human study shows asymmetric baseline-relative movement in the inferred index. Left-directed question selection produced the intended signed change in 20/21 sessions. Right push was effective for all nine initially Right-leaning participants but for only 12/29 initially Left-leaning participants. In the latter group, all observed endpoint nonresponders were identified using the first four adaptive responses. These findings establish the descriptive relationship between initial inferred position, directional response, and an early baseline-relative resistance flag in this experiment; they do not establish persuasion or changes in participants' underlying political beliefs.

VIII. Discussion and Conclusion

DTAG uses high-dimensional survey dependencies to select questions that systematically refine a predictive phenotype. The native experiments show directional movement using exclusively non-pole items; fixed-set replays indicate that selected question identities, rather than their precise order, account for much of the directionality. The index depends on the reference poles, candidate pool, and conditional model and should be interpreted as predictive position rather than a self-reported ideology score.

The expanded 71-session human-response study shows a strong asymmetry in directional responses: 20/21 Left-push sessions moved left, whereas 21/38 Right-push sessions moved right. Among initially Left-leaning respondents assigned Right push, 17/29 showed no net rightward movement from baseline after the full sequence. All 17 were identified from their first four responses, with two additional participants incorrectly flagged as resistant. Thus DTAG's observed directional response depends on the baseline predictive phenotype, and early trajectories can identify nonresponse to the chosen objective. The inferred shifts do not independently demonstrate changes in underlying political beliefs.

Acknowledgments

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TABLE IV

BASELINE-STRATIFIED DIRECTIONAL OUTCOMES AFTER THE FULL ADAPTIVE SEQUENCE. THE LAST COLUMN ALSO INCLUDES *CORRECT EARLY IDENTIFICATION OF RESISTANCE* FOR INITIALLY LEFT-LEANING PARTICIPANTS ASSIGNED RIGHT PUSH; IT IS NOT AN INTERVENTION-ONLY SUCCESS RATE.

Initial inferred phenotype	Left: direction achieved	Right: direction achieved	Right: movement or correct resistance
Initially Left	16/17 (94.1%)	12/29 (41.4%)	29/29 (100%)
Initially Right	4/4 (100%)	9/9 (100%)	9/9 (100%)
Overall	20/21 (95.2%)	21/38 (55.3%)	38/38 (100%)

TABLE V

HUMAN-STUDY RESULTS BY ASSIGNED POLICY. ENDPOINT DISPLACEMENT IS MEASURED FROM EACH PARTICIPANT'S SIX-QUESTION BASELINE; INTERVALS ARE 95% PARTICIPANT-BOOTSTRAP PERCENTILE INTERVALS (5,000 REPLICATES).

Policy	n	Baseline $I_\Phi < 0$	Baseline $I_\Phi > 0$	Mean $\Delta I_\Phi(20)$ (95% CI)	Directional success
Left	21	17	4	-0.0989 $[-0.1200, -0.0741]$	20/21 (95.2%)
Right	38	29	9	$+0.0225$ $[+0.0006, +0.0452]$	21/38 (55.3%)
Neutral	7	6	1	-0.0054 $[-0.0153, +0.0032]$	—
Random	5	2	3	-0.0045 $[-0.0512, +0.0413]$	—

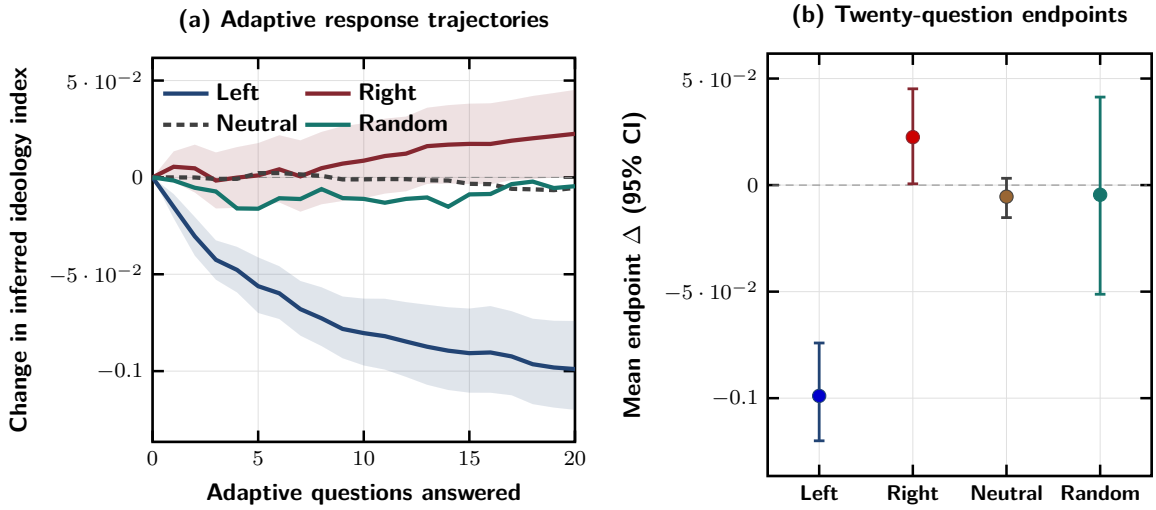


Fig. 3. **Updated human-study trajectories and endpoints (71 calibrated complete sessions).** (a) Mean baseline-relative change after each adaptive question in Left, Right, Neutral, and Random policies; the colored 95% participant-bootstrap confidence bands are shown for Left and Right. (b) Mean displacement after the twenty adaptive questions and 95% participant-bootstrap intervals, computed from complete archived sessions. All trajectories in this figure use only GCS-verified 20-answer completers, rather than imputing values for participants who stopped early.

TABLE VI

EARLY RESISTANCE CLASSIFICATION AMONG INITIALLY LEFT-LEANING PARTICIPANTS ASSIGNED RIGHT PUSH ($n = 29$). THE EARLY CLASSIFICATION USES THE FIRST FOUR ADAPTIVE QUESTIONS; THE OUTCOME IS NET MOVEMENT RELATIVE TO BASELINE AT THE END OF THE 20-QUESTION STUDY.

	No net rightward move	Moved rightward
Flagged resistant	17	2
Not flagged	0	10

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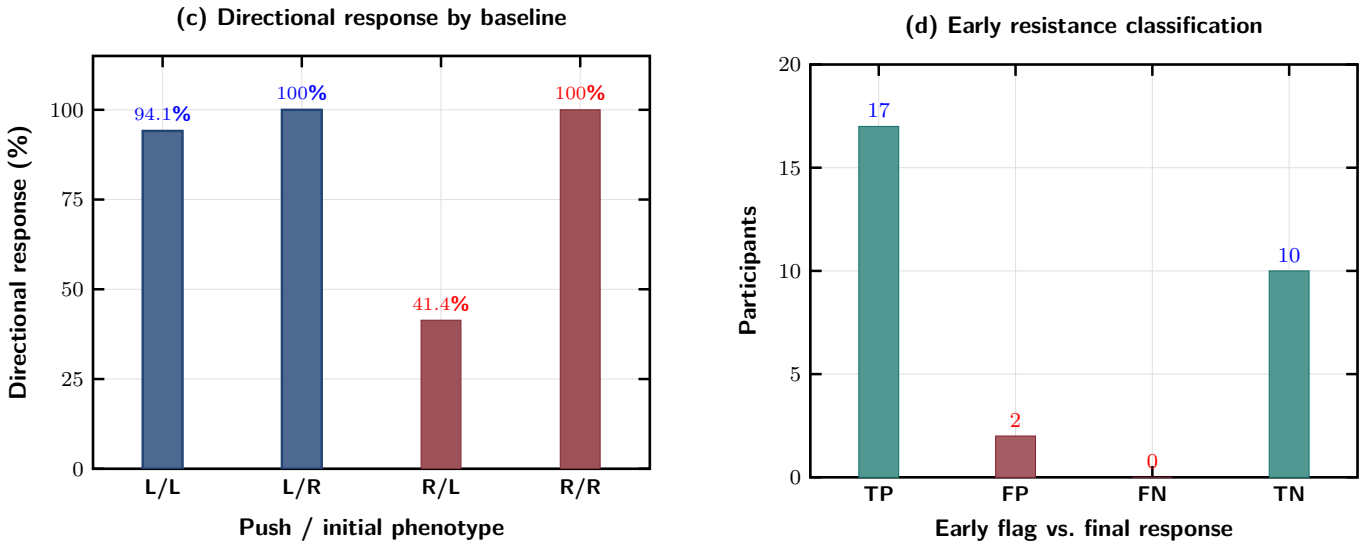


Fig. 4. **Direction-dependent response and early resistance detection.** (c) Proportion moving in the assigned direction after the study by initial phenotype sign and policy: L/L = 16/17, L/R = 4/4, R/L = 12/29, and R/R = 9/9, where the first symbol denotes policy and the second denotes baseline sign. (d) Classification from the first four questions of baseline-relative Right-push nonresponse at the end of the study among initially Left-leaning participants: 17 true positives (TP), 2 false positives (FP), no false negatives (FN), and 10 true negatives (TN).

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